

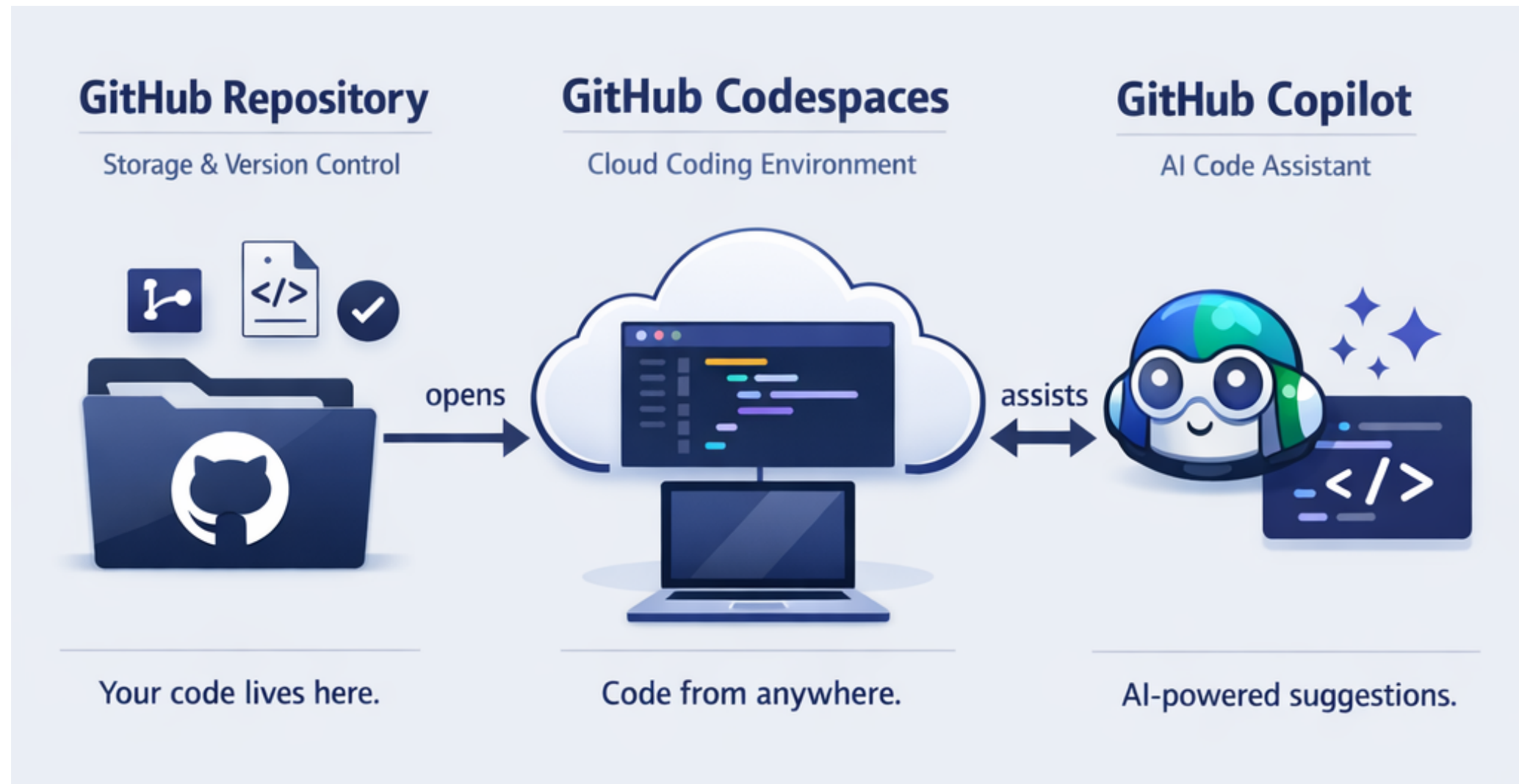


BAYES
BUSINESS SCHOOL
CITY ST GEORGE'S
UNIVERSITY OF LONDON

TOPICS LEARNED AND TAKEAWAYS



WEEK 1 ON GITHUB ECOSYSTEM



WEEK 1 ON PYTHON BASICS

- Primary **data types** in Python
 - `int` for integers (e.g., 1, 2, 3)
 - `float` for floating-point numbers (e.g., 3.14, -0.001)
 - `str` for strings (e.g., "Hello, World!")
 - `bool` for boolean values (`True` or `False`)
 - `list` for ordered collections of items (e.g., [1, 2, 3], ["apple", "banana", "cherry"])
- **Conditional and control flow** statements
 - `if-else` for branching logic
 - `for` loops for iterating over sequences
- **Indentation** is important in Python!

WEEK 1 & 2 ON MORE PYTHON BASICS

- Python uses **zero-based indexing**
 - so first element of a list is at index 0
- **nan** for representing *undefined, unrepresentable, or missing numerical values*
 - requires proper handling to avoid **disrupting** code execution
 - use **try-except** to catch and handle exceptions

WEEK 2 ON FIRST AUTOMATED FINANCIAL ANALYSIS

- **marimo** notebooks as replacement for Jupyter notebooks
 - with built-in *interactive features*
 - *automatic updates* of connected code cells
- Define function to compute **Altman Z-Score**
 - first with hypothetical hard-coded data
 - then with live company data (fetched from Yahoo Finance)
- Extend code to allow **user input** of any ticker symbol
 - **Visualize** Z-Score relative to Safe and Distress thresholds
- Fetch **panel data** of S&P500 firms via a **child marimo notebook**
 - **Save** fetched data to a **.csv** file for future access

WEEK 2 ON MARIMO DON'TS

- Do not re-define an object in another code cell
 - Variable names and function names must not be re-used
 - Or, will cause an error like this:

This cell wasn't run because it has errors

This cell redefines variables from other cells.

SP500_data was also defined by:

cell-75

Why can't I redefine variables?

marimo requires that each variable is defined in just one cell. This constraint enables reactive and reproducible execution, arbitrary cell reordering, seamless UI elements, execution as a script, and more.

Try merging this cell with the mentioned cells or wrapping it in a function. Alternatively, rename variables to make them private to this cell by prefixing them with an underscore (e.g. `_SP500_data`).

Learn more at our docs ↗: <https://docs.marimo.io/guides/reactivity/#multiple-definition-error>

TWO WEEKS INTO THE MODULE ...

- You learned how to
 - **automate the computation and visualization** of an important financial measure (Altman Z-Score)
 - using **panel data** of S&P 500 firms fetched from Yahoo Finance

WEEK 3 ON INTERACTIVE PLOTS

- Code templates for Plotly's
 - **Violin plot**
 - **Joyplot**
 - **3D scatter plot**
- Concepts of
 - first and third **quartiles** (Q1 and Q3), **median** (Q2), and **interquartile range** (IQR)
 - **lower and upper fences** for outlier detection (i.e., $1.5 \times \text{IQR}$ below Q1 or above Q3)
- To export marimo notebook as self-contained **html** file, use terminal command:
 - `marimo export html Wk03_InteractivePlots_Plotly.py -o Wk03_InteractivePlots_Plotly.html --sandbox --force`

WEEK 3 & 4 ON DATA PREPARATION AND CLEANING

- Pandas dataframes
 - **Read** data into dataframes
 - **Filter** rows for specific/missing values in a column
 - **Replace** missing values in a column with zero
 - **Create** new variables by applying functions to existing columns
 - e.g., creating lagged variables by shifting a column down by one row
 - **Group** data (e.g., by firm identifier)

WEEK 4 ON DATA ANALYSIS AND RESULTS PRESENTATION

- *Application example:*
 - The relationship between **Average cost of debt** (as a proxy for borrowing costs) and lagged **Z-Score** (as a proxy for credit risk)
- Code templates for
 - **Contingency table** by Median Split / Risk Zone
 - **Pooled regression**
 - **Risk zones box plot**
 - **2D scatter plot** with
 - interactive selection of observations by sector
 - reference lines and regression line

WEEK 4 ON MY FIRST WEB APP

- To deploy a dashboard app:
 - `marimo run Wk04w_Dashboard.py --sandbox`
- To publish a multi-tabbed webpage to the world:
 1. **Export** marimo notebook as a **html-wasm** file
 - `marimo export html-wasm Wk04x_MyPortfolio.py -o docs --sandbox --force`
 2. **Push** html-wasm and supporting files to **GitHub repo**
 3. Enable **GitHub Pages** for free hosting of webpages

FOUR WEEKS INTO THE MODULE ...

- You learned how to
 - **prepare and clean data** for analysis
 - conduct basic data analysis and present results with **interactive visualizations**
 - deploy a **dashboard app** and publish a **multi-tabbed webpage** to the world

WEEK 5 ON FREE AI RESOURCES

- Through a Moodle announcement post entitled *When you do not understand something, AI is your friend!*,
 - you were introduced to a variety of high-quality free AI resources beyond the GitHub Copilot sitting inside GitHub Codespaces:
 - GitHub Copilot (web interface with **multi-session history**)
 - supporting GPT-5 mini, Grok Code Fast 1, and other models
 - Qwen 3.5 Plus
 - Claude Sonnet 4.6
 - Gemini 3 Pro (Tools: **Guided Learning**); also, Gemini 3 Fast
 - Gemini 3.1 Pro Preview (on Google **AI Studio**)

WEEK 5 KEY TAKEAWAYS

- *AI is an integral part of your learning journey*
 - *for this module – and beyond*
- Don't wait for others to help you.
 - Engage in an *active learning* journey assisted by AI to *give yourself a brighter future!*

WEEK 6 ON DATA ANALYSIS WORKFLOW

- Following up on the discussion in previous weeks,
 - the whole pipeline of data analysis workflow was presented:
 1. **Data collection** (e.g., web scraping, APIs, databases)
 2. **Data preparation and cleaning**
(e.g., creating lagged variables, handling missing values)
 3. **Modeling development** (e.g., pooled OLS regression, panel data regression with year and industry fixed effects)
 4. **Analysis and Results**
(e.g., correlation analysis, regression results, robustness checks)
 - and illustrated with an example on the relationship between
 - **Tobin's q** (as a proxy for firm value) and **ESG score** (as a proxy for corporate sustainability performance)
 - controlling for other factors such as **firm size** and **leverage**

AGAIN, AI IS YOUR FRIEND

- You were encouraged to **seek AI assistance** in applying the data analysis workflow, such as by asking AI to:
 - **Explain** the concepts of Tobin's q and ESG score
 - **Provide examples** of Python code for implementing the tables in different steps of the workflow
- For example, you can
 - **upload the screenshot** of a descriptive statistics table to an AI assistant
 - ask it to **generate Python code for creating a similar table** based on your dataset
 - *Note:* Later in **Week 10**, the R and Python code for creating the tables in the Week 6 slides were provided to you

WEEK 6 ON DATA COLLECTION

- Structured data
 - Yahoo Finance has financial and stock data up to **last 5 years only** (unless you pay for premium access)
 - **WRDS** has financial and stock data dating back to 1960s:
 - **Compustat** and **CRSP** (subscribed by Bayes)
- Unstructured data
 - *Application example:*
Collecting company sustainability reports from corporate websites

WEEK 6 & 7 ON WEB SCRAPING

- **Challenges** in gathering unstructured data with web scraping:
 - Human verification triggered by **bot detection** blocks access to a website completely,
 - requiring careful handling of the scraping process to avoid detection
 - **Cookies consent banner** blocks access to the information on a webpage,
 - requiring identifying and clicking the *Accept* button to proceed
 - Locations of target data are often unknown in advance,
 - requiring comprehensive **webcrawling** to find the relevant information
 - Pdf files might not be text-searchable,
 - requiring **optical character recognition (OCR)** to extract text data from images

WEEK 7 KEY TAKEAWAYS

- Data science provides tools and techniques for *web scraping*, *webcrawling*, and *natural language processing* (NLP),
 - allowing collecting and analyzing unstructured data with **minimal human intervention**
- The **automation pipeline** can handle large volumes of data efficiently (e.g., multiple years of the S&P 500 companies' unstructured data),
 - **conserving** your time, energy, and attention **for more value-adding tasks** such as
 - learning from each identified page with a high frequency of the matched keywords (e.g., water pollution)

WEEK 8 INTRODUCTION TO LLMS

- A major risk of using **large language models (LLMs)** is **hallucination**
- Approaches to reducing the risk include:
 - **Hyperparameter tuning:** Making the model more deterministic (e.g., lower temperature)
 - **Prompt engineering:** Crafting prompts to provide guard rails towards finding the right answer, such as by asking the model to
 - **Think step by step** and
 - **Explain its reasoning** before giving the final answer
 - **Grounding:** Connecting the model to some source of truth through, e.g.,
 - Internet search
 - **Retrieval-Augmented Generation (RAG)**
 - **Few-shot examples**
 - **Using multiple models:** e.g., AI-as-judge

WEEK 9 API ACCESS TO LLMS

- Besides using LLMs through a chat interface,
 - accessing them through **application programming interfaces (APIs)** has the following key benefits:
 - **Automation** without manual intervention
 - **Integration** into other applications and workflows
 - **Scalability** for large volumes of requests (e.g., processing text data for all FTSE 100 companies across multiple years)
 - **Flexibility** to choose models and set hyperparameters
- *Application example:*
 - Determining the **sentiment** of forward-looking statements in FTSE 100 firms' annual reports

WEEK 8 & 9 KEY TAKEAWAYS

- LLMs are powerful tools for a variety of tasks
 - but **hallucination** is a major risk
- Techniques such as *prompt engineering* and *grounding* can be used to
 - guide LLMs towards more accurate and reliable outputs
- API access to LLMs enables the **automation** and **scalability**
 - required for large volumes of **judgment-based** analysis tasks, such as financial sentiment analysis

WEEK 10 ON FREE GPU ACCESS

- GPUs are the primary hardware powering modern AI
- They significantly accelerate computationally intensive tasks, such as
 - **Lemmatization in NLP:** Requires only 1/6th of the time by CPUs
 - *Application example: Bigram cloud analysis* of risk factors in *Magnificent 7* firms' annual reports
- Compared to GitHub Codespaces,
 - **Lightning.ai** provides even more generous free monthly credits sufficient for 75 hours of GPU usage
 - a GPU matching this compute power would require an incremental upfront investment of over £150

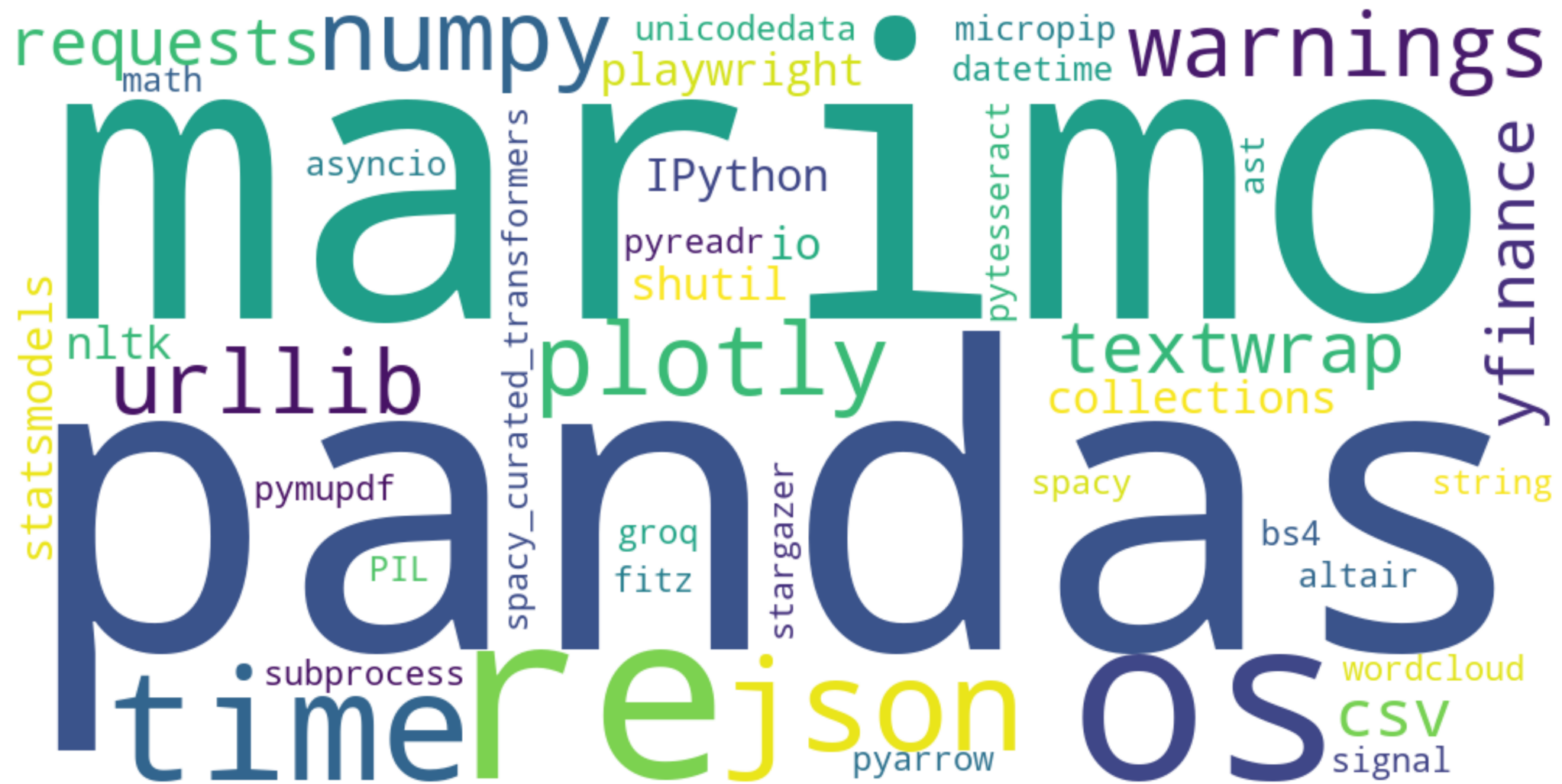
WEEK 10 ON R ECOSYSTEM

- R is a major programming language for data science, like Python
- R offers a **concise, expressive syntax**
 - avoiding Python's repetition of `lambda x:` and `x[" "]`
- R has a **rich ecosystem** of packages specialized for rigorous **statistical** modeling required to establish scientifically sound evidence
 - e.g., On the **DiD** website for **difference-in-differences** analysis, **over 15 packages in R** vs. **only 1 in Python**
- Unlike JSON-based Jupyter notebooks,
 - **R Markdown** notebooks are plain text files
 - **highly editable** in low-bandwidth environments
 - superior for **version control** and collaboration on platforms like GitHub

WEEK 10 KEY TAKEAWAYS

- *No single tool solves every problem*
 - Use Python for AI/Machine Learning (ML) tasks, but leverage R for rigorous statistical modeling
 - Selecting the “*right tool for the job*” is a key skill for data professionals
- The landscape of data science and AI **changes rapidly**
 - Continuous **self-learning and exploration** is the key to always choosing the right tool for the job

WEEK 1-10 PACKAGES USED



INDIVIDUAL PROJECT:

LAUNCHING YOUR PROFESSIONAL PORTFOLIO

- **The Core Goal:** Create the first draft of your personal portfolio webpage.
 - *The Long Game:* Polish this over the next 3 years so you have a professional, job-ready showcase by graduation.
- **Creative Freedom & Requirements:** You have total freedom to design your page, provided it:
 - Builds on the Week 4 code template.
 - Meets the assignment Guide requirements (Sections 3, 4, and 7.2).

INDIVIDUAL PROJECT: SUGGESTED WORKFLOW

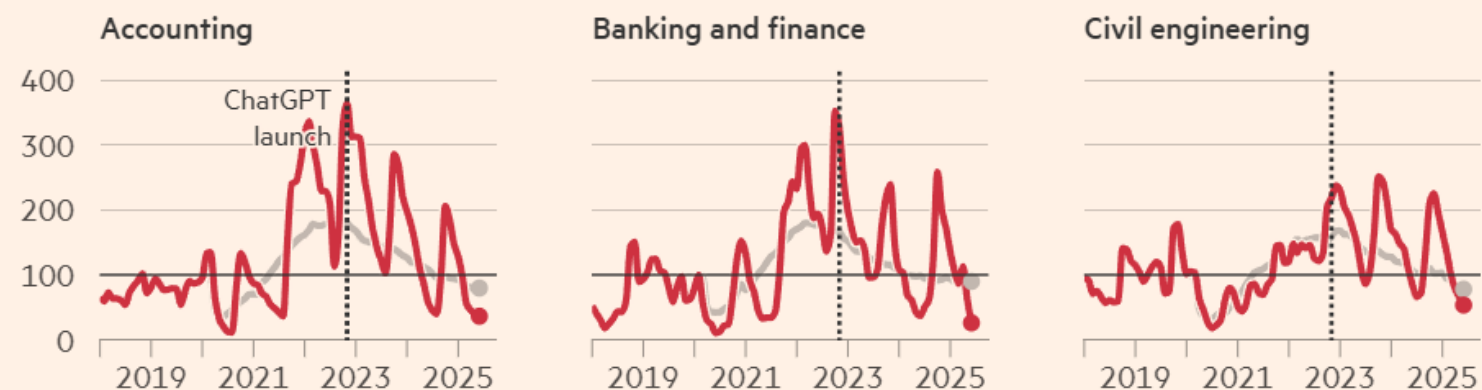
- **From Code to Global URL:**
 - *Start simple:* Swap “Jane Doe’s” CV for yours and customize your passions/interests.
 - *Publish:* Export as `html-wasm` via `marimo`, push to GitHub, and enable GitHub Pages.
 - *Result:* A live URL accessible on any device, no Python knowledge required!
- **Need Inspiration?** If you are stuck, step back from the code. Reflect on your strengths, what you care about, and the story you want to tell prospective employers.

AI AND DATA SCIENCE SKILLS

Graduate job postings are down across many UK sectors, even those less exposed to AI

Jobs advertised, monthly average (Feb 2020 = 100)

■ Total ■ Graduate



*In the age of AI, accountants are expected to have skills in areas like data analytics. [... a] survey noted that 58% of finance teams were using AI by 2024, mainly for anomaly and error detection, process **automation**, and analytics, but also found common hurdles such as “limited **data skills**” within firms ... proficiency in **programming language**, analytics tools, and cybersecurity are now **seen as crucial** for accountants to improve efficiency and provide strategic insights*

TO CONCLUDE

- This module was designed to equip you with the **data literacy** to
 - **adapt to** any new tool **the future** of finance and accounting throws at you

Helping you to **stand out** in the talent pool!

THANK YOU AND BEST WISHES!